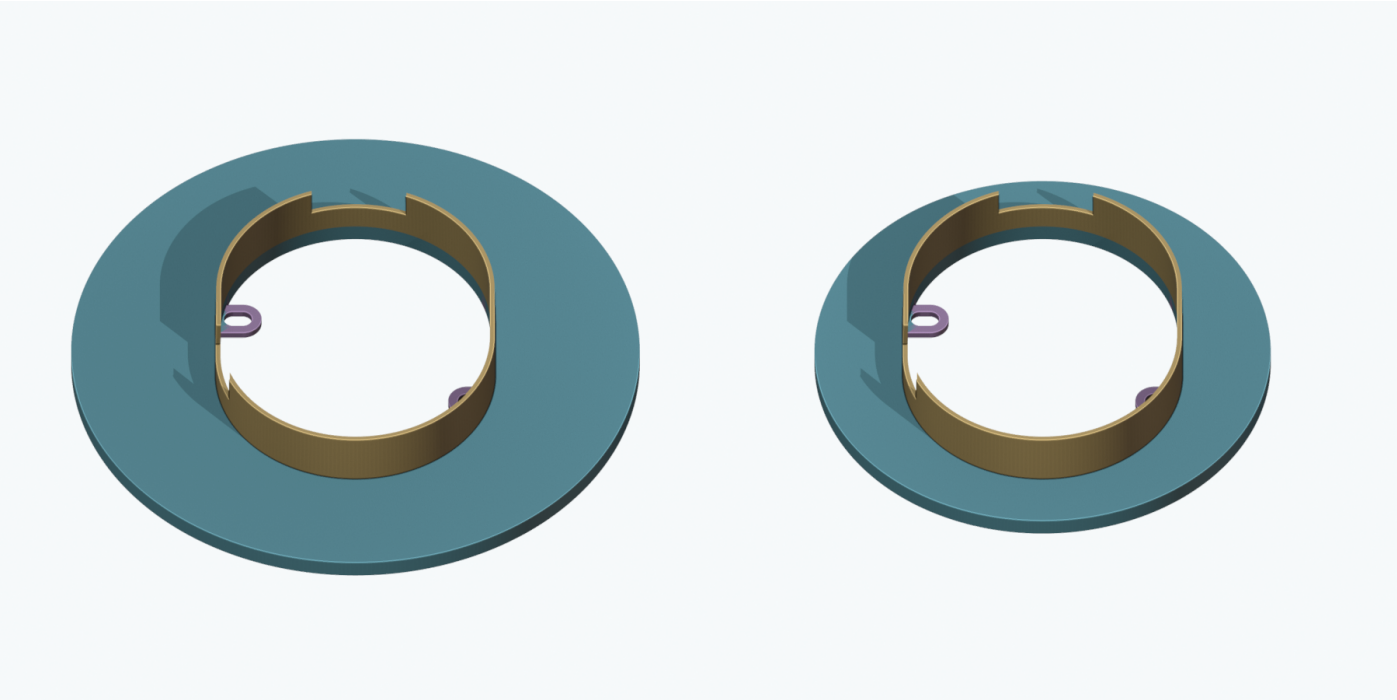
Two sizes. Same relieved collar.

RF04 small-bed alternate + unchanged RF03 original | October 17, 2026

Take both files to the printer. Choose one size and one format for each physical print. The smaller version only trims the outside face; it keeps both screw tabs and the complete collar.



Left: original 240 mm face. Right: smaller 192.55 mm face. Same scale.
Colors explain features; both are single-material PETG prints.

Choose a size	Original RF03	Small alternate RF04
Finished size, mm	240 x 240 x 32	192.55 x 192.55 x 32
Usable bed with 5 mm brim	250 x 250 mm	203 x 203 mm (rounded up)
Minimum face width outside collar	47.45 mm	23.725 mm: exactly half
Unchanged in both	8 mm face; 2 mm collar wall; 3 mm tabs	12 x 8 mm slots; 24 mm rear depth; first 12 mm continuous

Which file?

Small: SMALL_RF04 / RF04_SMALL_192_55_D24_RELIEVED_mm.3mf
Original: ORIGINAL_RF03 / PRINT_INSTEAD_RF03_D24_RELIEVED.3mf

Use the STL with the same name if the slicer cannot open the geometry-only 3MF. STL units are millimeters. STEP is an editable CAD alternative, not an additional part. Original RF03 files are byte-identical copies.

The original continuous 12 mm-collar design remains in the earlier RF03 package. This package contains the two relieved 24 mm versions requested here.

Print, locate, rehearse.

Temporary repair form - remove it after use

Slicer setting	Starting point
Material / profile	Unfilled PETG; use the friend's known filament and printer profile.
Scale / orientation	100%. Flat front face and tabs on bed; collar grows upward. Files already oriented.
Nozzle / layer	0.4 mm nozzle / 0.2 mm layers
Walls / solid skins / infill	5 walls / 1.2 mm top and bottom / 25% gyroid or cubic
Brim / supports / raft	5 mm / supports off after checking layer preview / raft off

At the printer

Confirm the preview shows one solid form at the listed size, lying flat. Use actual usable bed space after clips, exclusion zones and brim. A nominal 220 mm square bed can contain the small version if those exclusions allow it. Do not shrink the model to fit.

What the smaller face changes

The collar keeps the same central service opening clear and guides repair toward that boundary. The smaller face controls less of the surrounding repair surface. Beyond its edge, establish the repair plane against sound tile with a straightedge or suitable additional temporary form. Do not assume the smaller face bridges or supports the full damaged opening.

How the screws help

Remove the rod-end nuts for insertion. The unchanged slots provide coarse alignment with some play. They do not carry repair loads or prove exact centering. Independently support the face with broad padded bracing; do not tighten nuts to load the tabs or pull a distorted face flat.

Before any repair

Rehearse seating and removal on the real hardware. Check flat contact, clearance, stable bracing and withdrawal without trapping the form. The rear reliefs mean only the first 12 mm is continuous; keep repair within that continuous molding zone. Protect the service opening, slots and valve parts. Use a compatible supported repair, not filler spanning an unsupported void.

After suitable cure, remove all formwork, release film and pads. Neither version is a permanent escutcheon or waterproof liner. Existing no-tile-cutting and factory trim/seal requirements remain in force.

Material: the smaller CAD solid uses 47.0% less volume. Actual filament and time come from the slicer. The earlier $75 allowance covered one selected form plus a retry, not printing every alternative. Software checks verify files and retained geometry; physical fit, strength and water performance remain untested.